Shanghao Wu

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Education

Northeastern University Sep 2022 – Aug 2026
Bachelor of Science in Mathematics *GPA: 3.86/4.00*
Courses: Linear Algebra, Matrix Methods in Data Analysis, Probability & Statistics, Foundations of Data Science, Information Visualization, Databases, Mathematical Modeling & Methods, Real Analysis
Awards: Dean's List (5 semesters)

Skills

Programming: Python (proficient), SQL, HTML/CSS
Machine Learning: Linear Regression, Logistic Regression, Softmax Classification, SVM, KNN, K-Means, Random Forest, Gradient Boosting, PCA, KPCA, LDA, Regularization, Cross-Validation, Gradient Descent, Neural Networks (CNN, RNN), Kernel Methods, Markov Methods
Tools & Libraries: Git/GitHub, Scikit-learn, Pandas, NumPy, GeoPandas, Matplotlib, Jupyter, LaTeX

Experience

Northeastern University Boston, MA
Research Assistant under Prof. Lee-Peng Lee *Jan 2026 – Present*
- Implemented PCA, Kernel PCA with Gaussian kernel, and LDA from scratch in Python for face recognition on the 15-subject Yale Faces and 16,128-image Extended Yale B datasets
- Compared PCA, PCA+LDA, KPCA, and KPCA+LDA pipelines across varying training sizes; Fisherfaces achieved 100% accuracy on Yale Faces, outperforming PCA-only by up to 26.67%
- Demonstrated via kernel matrix analysis and synthetic data validation that KPCA offers no advantage over PCA on high-dimensional face data, with gains limited to inherently nonlinear, low-dimensional settings

Research Experience for Undergraduates (NSF-sponsored) Northeastern University
Research Intern *May 2025 – Jul 2025*
- Co-authored two combinatorics papers published on arXiv, contributing to the complete classification of cop numbers for all $n \times n$ classical chess graphs [arXiv: 2509.18516] and bounds on burning/cooling numbers for strong path products [arXiv: 2509.20572]
- Proved that two cops are insufficient on queen graphs Q_n for $n \geq 10$ by analyzing diagonal coverage limits and constructing a robber escape strategy based on a shortest-diagonal invariant Φ
- Led the liminal burning on paths component, establishing a lower bound for the critical number k^* using generating functions and combinatorial tiling arguments

Knack Northeastern University
Peer Tutor *Feb 2024 – May 2025*
- Tutored 50+ students one-on-one in Linear Algebra, Calculus, and Physics; designed targeted problem sets and walkthroughs to reinforce key concepts and improve exam performance

UKMT Remote
Mathematical Olympiad Grader / Lead Grader *Nov 2023 – Aug 2025*
- Graded and reviewed over 2,000 Olympiad scripts across BMO and JMO competitions; promoted to Lead Grader for BMO 2024
- Led and coordinated a grading team, refining rubrics and evaluation standards to improve inter-marker consistency and support transparent moderation

Publications

S. Ambrose, E. Angelone, J. Chen, D. Ma, A. Ortiz San Miguel, W. Watanabe, S. Whitcomb, **S. Wu**. "Cops and Robbers on Chess Graphs." *arXiv:2509.18516*, 2025.

S. Ambrose, E. Angelone, J. Chen, D. Ma, A. Ortiz San Miguel, W. Watanabe, S. Whitcomb, **S. Wu**. "Burning Games on Strong Path Products." *arXiv:2509.20572*, 2025.

Selected Projects

Diabetic Readmission Prediction | *Machine Learning* | *github.com/crushWu8/diabetic-readmission-prediction* Mar 2025 – Apr 2025
- Predicted 30-day hospital readmission for diabetic patients on the UCI Diabetes 130-US Hospitals dataset; built a reproducible ML pipeline with feature engineering, comparing Random Forest, Gradient Boosting, and KNN via cross-validation, achieving a positive-class F1 of 0.57
- Conducted fairness audits by age and gender; identified prior inpatient visits and high utilization as key readmission drivers

MBTA Bus Reliability Dashboard | *Data Visualization* | *github.com/crushWu8/mbta-bus-reliability* Nov 2025 – Dec 2025
- Cleaned and aggregated historical MBTA bus performance data, joining it with daily weather records to analyze on-time reliability across key routes
- Built and deployed an interactive dashboard on GitHub Pages visualizing reliability trends, precipitation impact, and route-level ridership shifts under varying weather conditions

FIFA Ticket Market Price Competition | *Mathematical Modeling* | *github.com/crushWu8/fifa-ticket-bertrand-duopoly* Nov 2025 – Dec 2025
- Modeled price competition between FIFA's official resale platform and the unofficial secondary market using a static Bertrand duopoly framework with a deviation parameter $\delta \in [0, 0.5]$
- Estimated a linear demand system from real transaction data, evaluated revenues under four pricing regimes, and visualized comparative statics of equilibrium deviations